Chandan S

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CAREER OBJECTIVE

An enthusiastic graduate eager to apply academic knowledge and practical skills to drive effective contributions to organizational growth. Possesses strong problem-solving abilities with a dedication to continuous learning and professional development.

WORK EXPERIENCE

ELECTRONICS ENGINEER - Neo San Private Limited – Nov 2025 - May 2026

- Performed functional testing and troubleshooting of electronic assemblies, improving product reliability.
- Identified and resolved hardware faults, reducing production delays.
- Supported manufacturing and quality teams in meeting production targets.
- Maintained quality documentation and ensured compliance with standard operating procedures.
- Collaborated with engineers to improve production efficiency and product quality.

EMBEDDED SYSTEM AND INTERNET OF THINGS - INTERN – ED GATE Technologies - Aug 2023 – Sept 2023

- Designed and developed a traffic light system using CSS Studio to regulate vehicular and pedestrian traffic flow at intersections and other road crossings.
- Conducted testing and validation to ensure functionality and reliability under various traffic conditions.
- Documented design decisions, processes, and technical specifications for future reference and maintenance.
- **Used Technologies**: Embedded C, CSS Studio, TM4C123GH6PM Microcontroller.

PROJECTS

LANDMINE DETECTION USING VOICE CONTROL ROBOT

- Integrated and configured the Speech Recognition Module with Arduino to capture and interpret voice commands in real-time.
- Conducted extensive testing and debugging to ensure the reliability and accuracy of voice recognition and command execution.
- **Used Technologies**: Speech Recognition Module, Arduino1.8.9, Embedded C.

LIGHT COMMUNICATION USING LI-FI TECHNOLOGY

- Designed and implemented an IoT device utilizing Li-Fi technology, where visible light serves as the medium for data transmission between two devices.
- Integrated sensors to enhance the device's functionality and data interaction capabilities.
- **Used Technologies**: Embedded C, Arduino IDE

COURSE

EMBEDDED SYSTEMS AND IOT – EMERTXE Technologies – Sept 2024 – May 2025

PROJECTS:

LSB IMAGE STEGANOGRAPHY

• Developed a LSB Image Steganography system using C programming to securely hide messages within images.
• Reducing processing time and memory overhead through efficient file I/O and pointer management.
• Enhanced data security and integrity by mastering byte-wise and bitwise manipulation for encryption and decryption.
• Implemented file handling operations for reading and writing image data without corruption.
• **Used Technologies**: Advanced C- pointers, bitwise operations, structures and file operations.
• **Key Challenges:** • Understanding of pixels and header of bitmap image file.
• Byte-wise and bitwise manipulation of data by declaring a local buffer to encode and decode.

INVERTED SEARCH

• Developed an inverted search system to create and manage word data bases from multiple text files, enabling efficient full-text search and document retrieval
• Implemented hashing and linked list for storing word occurrences, ensuring fast lookups.
• Designed a custom data structure to store word frequencies and their positions within documents.
• Key features include database creation, display, search, update and backup functionalities.
• **Used Technologies:** Data structure- hashing algorithms, text parsing and file operation.

CAR BLACK BOX

• Developed a Car Black Box system using a PIC microcontroller to record and analyze vehicle data for accident investigation.
• Designed an EEPROM-based data storage system for recording crash events and sensor data.
• Programmed the PIC microcontroller in Embedded C using MPLAB IDE and XC8 compiler.
• Implemented LCD display and buzzer alerts to notify drivers of system status and potential hazards.
• **Used Technologies**: PIC18F4580, CLCD, EEPROM, UART AND I2C.
• **Key Challenges:** Storing the events like gear change, speed and collision into EEPROM.

ADDRESS BOOK MANAGEMENT SYSTEM

• Designed and developed a command-line based Address Book application to store, manage, and retrieve contact information efficiently.
• Implemented key features such as add, update, delete, search, and sort for contacts using C programming.
• Utilized file handling in C to ensure persistent data storage across sessions.
• Applied structured programming concepts and dynamic memory allocation to enhance efficiency and flexibility.
• Incorporated input validation, modular code structure, and menu-driven interface for user-friendly interaction.
• **Used Technologies**: C, File Handling, Data Structures (Arrays, Pointers), Command Line Interface
• **Key Challenges**:
• Ensuring Data Persistence:
Challenge: Storing user data between program runs.
Solution: Implemented file handling to read/write contact information to a text file, allowing persistent storage.
•Handling Duplicate Entries and Input Validation:
Challenge: Avoiding redundancy and incorrect input formats.
Solution: Developed validation logic to detect duplicate entries based on name or phone number and sanitize user inputs.

EDUCATION

ST PHILOMENA SCHOOL – Bengaluru, India – **2018**

- SSLC

RNS PRE UNIVERSITY - Bengaluru, India – **2018 – 2020**

- PU(PCMCS)

DON BOSCO INSTITUTE OF TECHNOLOGY - Bengaluru, India – **2020 - 2024**

- BE - Electronics and communication Engineering

SKILLS

- Microcontroller (PIC 8051)
- Linux
- C Programming
- C++
- SQL
- HTML
- Verilog HDL
- Embedded C
- Matlab

SOFT SKILLS

- Problem Solving
- Analytical
- Leadership